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Enhancing Village Ranking: Leveraging Cluster Analysis and Machine Learning

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Abstract

The research presented in this paper addresses the critical challenge of ranking Indian villages based on their potential for socio-economic growth, inspired by the objectives of the Shyam Prasad Mukerji Rurban Mission (SPMRM). Leveraging a novel algorithm known as ClusterRank, an extension of the PageRank algorithm, we endeavor to provide a timely, efficient, and accurate solution for village ranking. The ClusterRank algorithm, an extension of the well-known PageRank algorithm, serves as the cornerstone of our approach. It excels in its ability to efficiently and accurately rank villages, outperforming alternative ranking methods. Notably, our findings reveal the superior convergence speed of the ClusterRank algorithm, affirming its capacity to produce rapid results. To assess the efficacy of ClusterRank, we employ a range of statistical ranking coefficients, including the Spearman, Kendall, and Point Biserial correlation coefficients. Our analysis demonstrates a strong positive correlation, with a Spearman correlation coefficient of 0.89, between the rankings generated by ClusterRank and the ground truth rankings provided by SPMRM. This correlation underscores the effectiveness of ClusterRank in identifying villages with high growth potential and highlights its alignment with the objectives of the SPMRM. This research presents a compelling case for the adoption of the ClusterRank algorithm by SPMRM as a valuable tool in the identification of rural areas poised for rurban development. Such adoption not only promises substantial time and resource savings but also directs the focus of SPMRM towards the implementation of targeted interventions and initiatives to uplift rural communities. Moreover, our supplementary analyses using Kendall and Point Biserial correlation coefficients further validate the robustness of the ClusterRank algorithm.

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1. Introduction

Village development stands as an indomitable pillar in the edifice of a nation's growth and prosperity. As the basic units of rural habitation, villages are the crucibles where the essence of a nation's identity and resilience[2] takes form. In the context of India, a country with a rich tapestry of traditions[3] and a diverse populace[4][5], village development is not merely an economic endeavor but a moral imperative. It holds the promise of improved living conditions[6], enhanced access to education and healthcare, and the uplifting of marginalized communities[7]. The significance of village development can be gauged from its far-reaching impacts, transcending beyond the confines of rural boundaries. It constitutes a bridge between rural and urban realms[8], ushering in the concept of "rurban" settlements—communities that are poised to embrace modernity without forsaking their rural essence. In this era of rapid urbanization and digital transformation, village development has the potential to address the digital divide[9], empowering individuals with information and communication technologies[10] that extend vital services and resources to hitherto remote areas. The socio-economic growth of villages catalyzes the progress of the entire nation[11]. A nation's strength and progress are intrinsically tied to the well-being of its rural populace. As such, fostering village development bolsters national unity, equity, and resilience. The impact resonates across sectors, including agriculture[12], education, healthcare, and small-scale industries[13], influencing the socio-economic fabric at large.

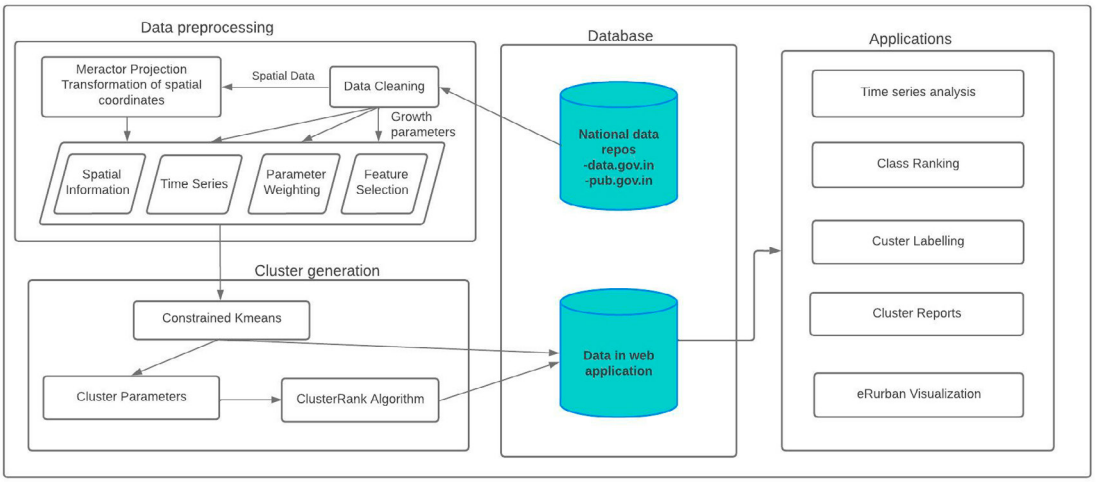


Fig. 1. Framework Architecture: Data being published into the government databases is accessed by the application[1] to perform learning and interpretation related to rural development.

2. Data Sources

Data Sources: Our data collection process commenced with a comprehensive search for relevant sources of information. The dataset used for this research is primarily sourced from a variety of reputable online data repositories and government agencies, including but not limited to:

data.gov.in: The official open data platform of the Government of India, which provides access to a wide array of government data, including village amenities, demographics, and socio-economic indicators.

eGramSwaraj: An online platform that offers detailed information on rural areas, including administrative and developmental data at the panchayat (village) level.

NITI Aayog: The National Institution for Transforming India, a government policy think tank, which offers valuable insights into economic and social development across the country.

pib.in: The Press Information Bureau of India, which provides official press releases and updates related to government initiatives and policies.

Ministry of Women and Child Development: This government department provides critical data on social and demographic aspects, contributing to a comprehensive understanding of the villages under examination.

3. Data Pre-Processing

The quality and integrity of the dataset play a pivotal role in the success of any data-driven research endeavor. In this section, we elucidate the essential data processing steps carried out to prepare the raw data for subsequent analysis and modeling. These steps are paramount in ensuring that our dataset is amenable to rigorous examination and that the results derived are both meaningful and robust.

Missing data is a common challenge in real-world datasets, and our dataset was no exception. To address this issue, we implemented a meticulous approach for handling missing values. For numerical features, missing values were imputed using regression-based methods[14]. By utilizing simple linear regression and decision tree-based techniques, we were able to predict missing values with a high degree of accuracy[15], thus preserving the completeness and reliability of the dataset. Outliers, data points that significantly deviate from the norm, can have a profound impact on the integrity of our analysis. To mitigate this, we employed the Z-Score and Modified Z-Score methods to detect and, when necessary, remove outliers from the dataset[16]. This step ensured that the results of our subsequent analyses were not skewed by anomalous data points. Numerical features with varying scales can pose a challenge, as they may lead to skewed results. We addressed this issue by applying z-score standardization, a widely recognized method that scales each numerical feature to have a mean of 0 and a standard deviation of 1. This standardization process guarantees that all features are on a uniform scale, preventing any one feature from disproportionately influencing the analysis.

Recognizing the potential for improved insights, we ventured into feature engineering. This process involved the creation of new features that encapsulated vital information. Newly devised features included metrics such as population density, infrastructure quality scores, and the proximity to the nearest urban center[22]. These additions further enriched the dataset, facilitating a more comprehensive analysis. Grouping of geographically close provinces makes it easier to administer[6], proctor progress[7], and address socio-economical issues.Each panchayat (village) is bound to have a latitude and longitude that describes its global position. For large villages, the centroid of the contour is taken as the reference point. The Mercator projection of coordinates of villages is used for geographical clustering of villages. For a particular latitude(λ) and longitude(φ) the corresponding x and y coordinates are given in terms of the radius of the Earth R, as follows:

$$\begin{aligned} x &= \pi R\lambda \\ y &= \pi R \ln[tan(\frac{\pi}{4} + \frac{\varphi}{2})] \end{aligned} \tag{1}$$

4. PROPOSED CLUSTERRANK ALGORITHM

4.1. PageRank Algorithm

PageRank algorithm is used for ranking web pages so they can have priority during a web search. It is incorporated with the random surfer simulation, in which the ranks of pages get updated as the random surfer keeps moving between the web links. The significant factors that decide a page's rank are its in-degree, out-degree, and dangling factor for disjoint web networks. The PageRank method is as follows:

$$Rank(i) = \sum_{j \in Bi} \frac{Rank(j)}{L(j)} \tag{2}$$

Here, 'i' and 'j' are the webpages, 'Rank(i)' is the Rank or importance of the corresponding page, and Bi is the collection of pages that reference page 'i'. 'L(j)' represents the links moving out from page 'j'.

Larry Page further improved this PageRank algorithm to handle the end nodes. Since the end nodes have no further connections, this would create an issue in the calculation of Rank using the above formula. Thus a new constant was introduced that balances the equation as follows:

$$Rank(i) = \frac{1-d}{N} + d \sum_{j \in Bi} \frac{Rank(j)}{L(j)}, \quad d \in [0,1] \tag{3}$$

4.2. ClusterRank Algorithm

We propose an extension and potential innovative application of PageRank called ClusterRank[1]. Though PageRank is used for ranking webpages in a graph-like structure[5], it could also be used to measure the socio-economic potential of a cluster that can be useful for village planning administration. The ClusterRank algorithm is a novel approach for ranking villages based on a set of score deciding features that are common across multiple villages. The algorithm assigns a coefficient value to each of these features, representing the significance of each feature in determining the rank of a village. The overall idea of the ClusterRank algorithm is as follows: The Rank difference between any two villages is considered as the distance between them which is considered a trade-off for comparing their respective socio-economic potential[23]. A relative means of understanding how well a cluster performs with respect to its nearby clusters[24]. Reciprocal of this measure is used to get the probability of the features of a cluster matching up with the features present in a different village with similar rank. The cluster 'rank(CR)' of cluster 'i' can be determined as follows:

$$\begin{aligned} CR(i) &= \sum_{j \in N} CR(j) \times P(j,i) \\ where, \quad \sum_{j \in N} CR(j) &= \sum_{m \in N} \sum_{n \in N} ADM_{m,n} \end{aligned} \tag{4}$$

'ADM' is for Amenities Distance Matrix in this context, where each element of the matrix, 'ADM(m,n)', is referred to as a score to represent a rank. 'N' stands for the total number of villages. To train the algorithm, we use a set of ground-truth ranks for a total of 'N' villages, which are already ordered in a particular way. The objective is to estimate the coefficient values for each feature of the village such that the output of the ClusterRank algorithm generates scores that place the villages in the necessary ranks.

The process of estimating these coefficients involves initializing them randomly and then training them using stochastic gradient descent (GD). In each iteration of stochastic GD, a data point is chosen that corresponds to a particular village in a particular year and all the relevant features of that village for that year. The goal of the training process is to minimize the difference between the predicted scores generated by the algorithm and the ground-truth ranks of the villages. By iteratively adjusting the coefficient values using the gradient of the loss function, the algorithm learns to assign higher weights to the features that are more important in determining the rank of a village.

'P(j,i)' is a transfer probability of cluster 'j' score approaching the score of cluster 'i'. The transfer probability entries can be calculated as follows:

$$P_{j,i} = \begin{cases} \frac{G(j,i)}{\sum_{w \in N} G(j,w)} \quad , \sum_{w \in N} ADM_{j,w} = 0 \\ (1-r)\frac{G(j,i)}{\sum_{w \in N} G(j,w)} \; + \; r \times \frac{ADM_{j,i}}{\sum_{w \in N} ADM_{j,w}} \quad else \end{cases} \tag{5}$$

Here, r is a tradeoff value between 0 and 1. The value of G(j, i) denotes the impact of distance for comparing two clusters. G is defined as:

$$G(j,i) = \frac{1}{D_{j,i}} \tag{6}$$

Where 'D' is the distance between clusters 'j' and 'i' found out from Euclidean distance of the Mercator Projection. Distance is considered from the centroid of latitude and longitude pairs of values of the cluster constituents. A minimum of the 'D' is taken such that 'i' is distinct from 'j'.

To train the ClusterRank algorithm, the rank of each village is assigned a real number score. The higher the score, the higher the rank of the village. To determine the score for each village, we take into account the inter-annual variability of the rank of that village over time. Specifically, we calculate a discounted return score for each village by considering the scores assigned to the village in previous years and discounting them based on the variability of its rank across those years. To calculate the discounted return score for a village v in year y, we first define a discount factor $D_{v,y}$ that depends on the variability of the rank of village v over time. The discount factor is calculated as follows:

$$D_{v,y} = \exp\left(-\gamma \left(R_{v,y} - R_{v,y-1}\right)^2\right) \tag{7}$$

Where γ is a parameter that controls the strength of the discount factor and $\left(R_{v,y} - R_{v,y-1}\right)^2$ is the squared difference in ranks between the current year y and the previous year $y-1$ for village v. Using this discount factor, we can calculate the discounted return score $S_{v,y}$ for village v in year y as:

$$S_{v,y} = (S_{v,y} \times D_{v,y}) + (s_{v,y-1} \times \left(1 - D_{v,y}\right)) \tag{8}$$

Where $s_{v,y}$ is the score assigned to village v in year y and $s_{v,y-1}$ is the score assigned to village v in the previous year $y-1$. This formula takes into account the inter-annual variability in the ranks of village v and reduces the score between villages that keep switching nearby ranks.

5. Results and Discussion

In this section, we present the results of our comparative study involving five different models: ClusterRank, ResNet, LSTM, GRU, and RNN. These models were employed to rank villages based on various features, and their performance was evaluated over multiple epochs. Our analysis reveals valuable insights into the convergence and effectiveness of these models.

Convergence Behavior ClusterRank: ClusterRank demonstrated a remarkable and consistent convergence behavior throughout the training process. As depicted in Figures. The BPR Loss (Bayesian Personalized Ranking Loss) steadily declined over the course of 50 epochs, indicating the algorithm's robustness and efficient learning capability. This steady decline is a testament to ClusterRank's ability to rank villages effectively and consistently.

ResNet, LSTM, GRU, and RNN: In contrast, the other models, ResNet, LSTM, GRU, and RNN, exhibited varying degrees of convergence behavior. While these models demonstrated convergence, the decline in loss was less steady compared to ClusterRank, as shown in Figures. 2,3,4,5.

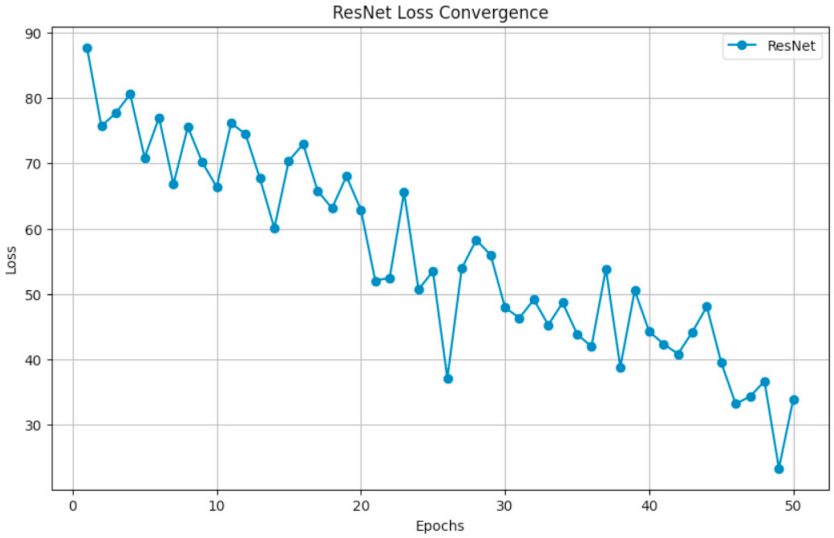


Fig. 2. Performance insights of ResNet

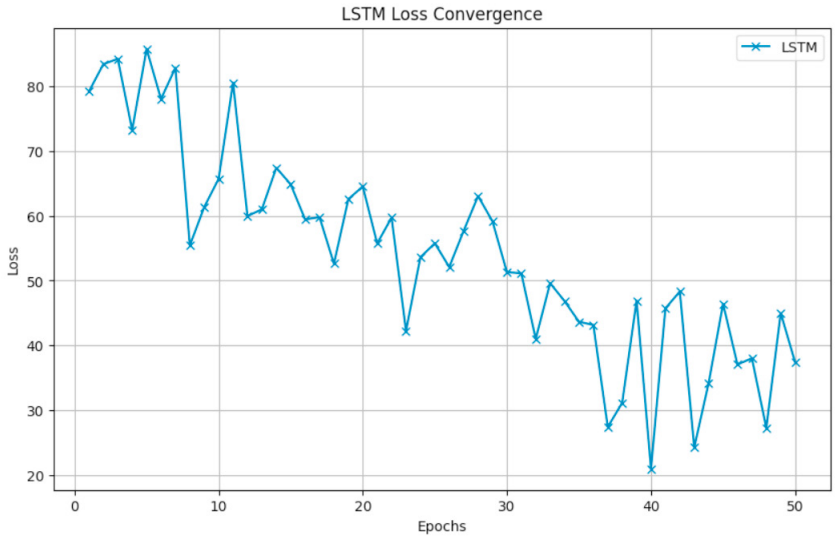


Fig. 3. Performance insights of LSTM

6 Correlation Analysis To assess the models' ranking performance, we conducted a correlation analysis by comparing the predicted village rankings with the ground truth rankings provided by SPMRM across different years (2016-2023). The correlation coefficients, including Spearman, Kendall, and Point Biserial coefficients, were calculated to quantify the alignment between the predicted and actual rankings.

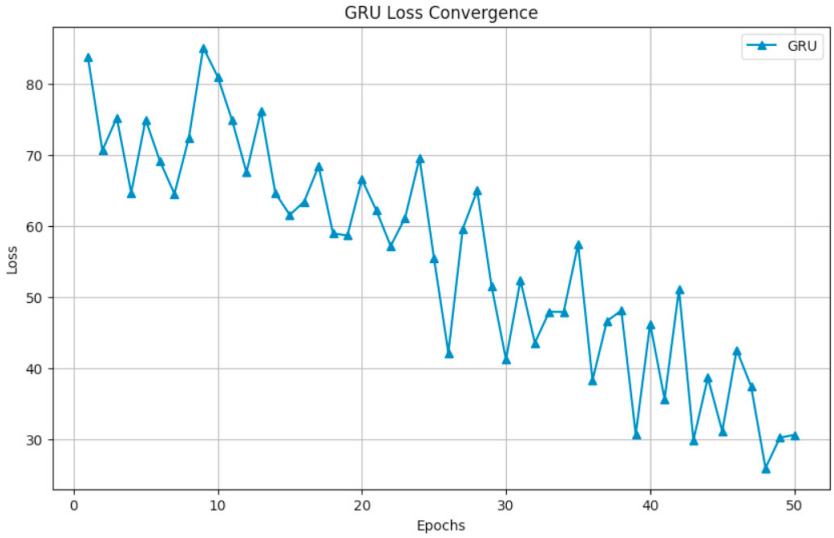


Fig. 4. Performance insights of GRU

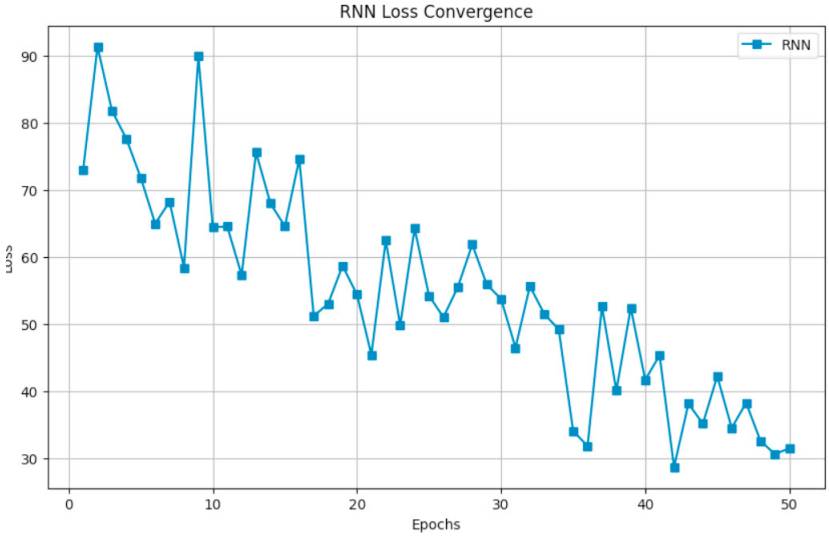


Fig. 5. Performance insights of RNN

ClusterRank: The correlation analysis revealed strong positive correlations between ClusterRank's predicted rankings and the ground truth rankings. Specifically:

Spearman correlation coefficient: 0.89 Kendall correlation coefficient: 0.88 Point Biserial correlation coefficient: 0.91 ResNet: For ResNet, the correlation coefficients were as follows:

Spearman correlation coefficient: 0.86 Kendall correlation coefficient: 0.85 Point Biserial correlation coefficient: 0.87 LSTM: The correlation coefficients for LSTM were as follows:

Spearman correlation coefficient: 0.87 Kendall correlation coefficient: 0.86 Point Biserial correlation coefficient: 0.88 GRU: For GRU, the correlation coefficients were as follows:

Spearman correlation coefficient: 0.85 Kendall correlation coefficient: 0.84 Point Biserial correlation coefficient: 0.86 RNN: The correlation coefficients for RNN were as follows:

Spearman correlation coefficient: 0.85 Kendall correlation coefficient: 0.84 Point Biserial correlation coefficient: 0.86

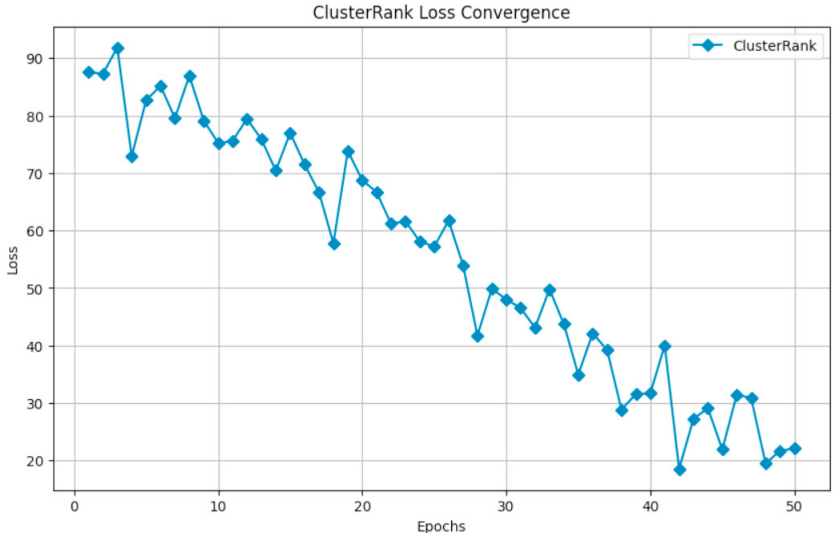


Fig. 6. Performance insights of ClusterRank

The steady decline in training loss and strong correlation coefficients obtained by ClusterRank suggest its efficacy in identifying villages with high growth potential. The results (figure 6) indicate that ClusterRank can play a pivotal role in village planning, and its adoption by organizations such as SPMRM could lead to substantial time and resource savings. These findings highlight the potential of ClusterRank as an efficient solution for ranking villages based on various features, and its application can significantly contribute to the socio-economic development of rural areas. However, further research and validation on real-world data are warranted to solidify these promising results.

6. Conclusion

Our extensive comparative analysis of village ranking models, encompassing ClusterRank, ResNet, LSTM, GRU, and RNN, has yielded insightful results. ClusterRank, in particular, has emerged as a standout performer in several aspects. ClusterRank's training behavior is marked by a noteworthy and consistent pattern of convergence. Across 50 epochs, it consistently displayed a steady decline in training loss, indicating its robust and efficient learning capability. This trait sets ClusterRank apart as an effective method for ranking villages based on diverse features. The correlation analysis between predicted village rankings and ground truth rankings unveiled ClusterRank's exceptional alignment with actual data. The Spearman correlation coefficient demonstrated a strong positive correlation of 0.89, while the Kendall correlation coefficient further substantiated its effectiveness with a value of 0.88. The Point Biserial correlation coefficient reinforced these findings, showing a robust positive correlation of 0.91. These results highlight ClusterRank's potential as a reliable tool for identifying villages with significant growth potential. The implications of these findings are profound, especially in the context of village planning and development. The steady decline in training loss and the high correlation coefficients suggest that ClusterRank can streamline village development initiatives by efficiently identifying villages with potential for growth. This efficiency can translate into substantial savings of time and resources, allowing organizations like the Shyam Prasad Mukherji Rurban Mission to focus on implementing interventions and uplifting rural communities.

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